



SEYED MOHAMMAD SALEHI

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SUMMARY

I am a dedicated engineer pursuing an M.Sc. in Computer Vision, Robotics, and Machine Learning at the University of Surrey, with a B.Sc. in Electrical Engineering from Shahid Beheshti University. My academic and research interests lie in robotics, Artificial Intelligence, and intelligent autonomous systems. I have taught robotics and AI for two plus years in a company and at my university, and I have written a research article, deploying advanced deep learning algorithms. I aim to contribute to cutting-edge research and projects that deploy AI and robotics in real-world applications.

WORK EXPERIENCE

Electrical Control Engineer, Rhizo PTX (iTEK Project) Current Project

- Designed and integrated a hardware–software dashboard for system-wide PID control and layered safety logic; engineered two transistor-based switching stages to regulate system voltage and current, then implemented their controls and interface buttons in the dashboard software.

Teaching Assistant, Faculty of EE, Shahid Beheshti University February 2024 – July 2024

- Microprocessors and Computer Architecture, led by Professor Nader Kazemi.

Teaching Assistant, Faculty of CE, Shahid Beheshti University February 2023 – February 2024

- Artificial Intelligence, led by Professors Shahabedin Nabavi and Atefe Aghaei.

Software Engineering Team Member, Sarmad Company February 2023 - July 2023

- Developed embedded-system software in C using Visual Studio and Code Composer Studio.

Robotics Training Expert, Sigma Company April 2020 - September 2022

- Taught robotics and coached students for national and international competitions; co-developed the curriculum, robot bodies, electronic circuits, and initial board firmware for training packages. Helped organize robotics competitions at the University of Tehran (2021) and in India (2022).

Software Engineering Team Member, Cube Electronic Company April 2022 - June 2022

- Developed C/C++ firmware for STM32 microcontrollers using STM32Cube and Keil MDK.

EDUCATION

M.Sc. in Computer Vision, Robotics, and Machine Learning September 2025 - September 2026

University of Surrey

- Thesis: Human Motion Prediction for Safe and Trustworthy Autonomous Navigation in Urban Environments**
Evaluating DeMo, SEAM, and SHARP on Argoverse 2; extending SHARP with Mamba state-space encoders and controlled attention ablations for accuracy, temporal modelling, and deployment.
- Finalised-module average: **71.75%** (Applied Machine Learning 85; Research, Professionalism & Innovation 75; Medical Robotics 67; Advanced Computer Vision & Deep Learning 60).
- Current non-zero coursework: Computer Vision & Pattern Recognition 100; Speech Processing 80; Speech Recognition 98; Artificial Intelligence of Things 73. Unfinished examination marks omitted.

B.Sc. in Electrical Engineering September 2019 - July 2023

Shahid Beheshti University

- Selected results: English 100, Energy Systems Analysis 98.5, Technical English 97.8, Microprocessor Laboratory 97.5, Programming 97, Embedded Systems 96.3, Control 91.
- B.Sc. thesis: Simulation of a Self-Driving Car (92.5/100); GPA: 77.0/100.

High School Diploma in Mathematics & Physics October 2016 - July 2019

National Organization for Development of Exceptional Talents (Helli 4 School)

- GPA: 96.0/100

PROFESSIONAL ROLES

- **Treasurer of the University of Surrey Computer Science Society:** We run academic and social events monthly, as well as larger events throughout the year. (March 2026-March 2027)
- **Treasurer of the University of Surrey Languages Society :** Whether speaking in English or a language you are familiar with, our society engages in a wide range of activities, from foreign film nights to pub quizzes. (March 2026-March 2027)
- **Treasurer of the University of Surrey Walking Society:** We offer relaxing social walks around Guildford, serving as a community-focused group to help students explore the local area and meet new people. (March 2026-March 2027)
- **Course Representative of the Computer Vision, Robotics, and Machine Learning Program at the University of Surrey:** We provide a way for students to communicate their opinions and experiences to the staff who deliver their courses, and to make change happen (Sept 2025-Sept 2026)
- **University of Surrey UAS Team Peryton — IMechE UAS Challenge 2026 Participant**
Built and flew a UAV as a multidisciplinary team; my primary contributions were flight-control programming, parameter tuning, and electronic-circuit design and fabrication. [Certificate of participation](#)
- **University of Surrey Stage Crew (USSC) Member :** We prepare the sounds and visuals of the stage for any events (Sept 2025-Sept 2026)
- **University of Surrey Electronics & Amateur Radio Society (EARS) Member :** We are preparing our robot for the [Battlebots](#) Robotics Competition this year (Sept 2025-Sept 2026)
- **University of Surrey Sports Park (SSP) Volunteer :** I am part of the [Halow Project](#), which helps adults with learning disabilities (Sept 2025-Sept 2026)
- **Electrical Engineering Scientific Association Magazine (Kahroba-ye-Beheshti) Chief Editor :** Our (2020/21) edition was a candidate for the best scientific magazine in EE(Sept 2020-Sept 2021). Also, **Co-author :** Where I wrote about the IoT in the (2019-2020) edition of the magazine
- **Electrical Engineering Scientific Association Research Secretary :**
We participated in the International Harekat Festival, where we demonstrated all of our achievements in the Electrical Engineering Faculty (Sept 2020-Sept 2021)

RESEARCH INTERESTS

- Artificial Intelligence, Machine & Deep Learning, Embedded Systems & IoT, Robotics & Autonomous Systems, ASIC & FPGA, Image & Video & Speech Processing

CERTIFICATES

- IMechE UAS Challenge 2026 Certificate of Participation, [Halow Project Volunteer Certificate](#), Machine Learning Specialization (Stanford Uni./Coursera), Mathematics for Machine Learning and Data Science Specialization (DeepLearning.AI/Coursera), Self-Driving Car Engineer Training (Udacity), Essay Writing Skills (Amirkabir Uni.), English Language Course (ILI, 12 years), English TTC (Kish Language Inst.), IELTS English Teacher Training Certificate (Shahid Beheshti Uni.), Preliminary and Advanced Photoshop (Shahid Beheshti Uni.), Photoshop Skill Certificate ([TVTO](#)). Originals on LinkedIn.

PUBLICATIONS

- Enhancing SOC and SOH Estimation in Lithium-Ion Batteries through SABRE and RELIANT-Based Approach: I used advanced DL models and architectures, enhancing the prediction of battery states

ENGLISH TESTS

- GRE (General test): Test date: 09/01/2024 / I am also learning German, and I am at the A1 level now
- IELTS (Academic): 7.0 (overall score)L: 7.5, W: 6.5, S: 6.5, R: 6.5, Test date: 17/11/2024
- IELTS (Academic): 7.0 (overall score)L: 7.5, W: 6.5, S: 6.5, R: 8.0, Test date: 12/05/2022

TECHNICAL SKILLS

- Circuit Design (Advanced Design System, PrimeSim HSpice, Cadence PSpice, Proteus Design Suite), Circuit Synthesis (Vivado Design Suite, Xilinx ISE Design Suite), Deep Learning (Conda & JupyterLab & Colab IDEs), C/C++ Programming (STM32Cube, Keil MDK, Arduino, Atmel Studio AVR, Visual Studio, Code Composer Studio), IoT(COOJA), Bibliographic Database (EndNote & Overleaf), 3D CAD (Autodesk Fusion), Math (Mathworks Matlab/Simulink)

PROFESSIONAL REFERENCES

- **Professor Kevin Wells** — MSc Thesis Supervisor; Professor of AI in Human and Veterinary Healthcare, University of Surrey (CVSSP). k.wells@surrey.ac.uk
- **Dr Gareth Beeston** — Rhizo PTX/iTEK Project Supervisor; Senior Project Officer, iTEK, Surrey Student Enterprise, Innovation and Impact. g.beeston@surrey.ac.uk; [profile](#)